# Ramesh Radhakrishnan

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## Research Profile

Theoretical physicist specializing in gravitational perturbations, gauge-invariant wave observables, rotating wormhole geometries, and spectral methods in curved spacetime. Published work spans gravitational-wave polarization in general relativity and metric $f(R)$ gravity, geometric invariants, nonseparable rotating-wormhole spectra, quantum-gravity models, and quantum-vacuum particle creation. Current research develops exceptional-point (EP) methods for non-Hermitian spectra, quasinormal-mode (QNM) degeneracies, resolvent residues, and source-driven black-hole response. Future research aims to investigate whether EP phenomena can influence extreme-mass-ratio inspiral (EMRI) excitation and gravitational-wave spectroscopy; gravitational spin-2 and observable waveform extensions remain proposed rather than established results.

## Education

**Ph.D., Theoretical Physics**, Dept of Physics & Astronomy, Baylor University. GPA-3.85 May 2026
*Dissertation: Perturbations and Symmetry in Gravitational Theories: Wormholes, Wave Dynamics, and Quantum Effects in Rotating Spacetimes*
**Advisor:** Dr. Gerald Cleaver
**M.A., Theoretical Physics**, Baylor University 2022
**M.S., Physics**, University of Texas at Dallas 2011
**M.S., Computer Science**, Old Dominion University 1993
**A.M.I.E., Mechanical Engineering**, Institution of Engineers (India) 1988

## Research Interests

- Black-Hole Perturbation Theory and QNMs
- Exceptional Points and Non-Hermitian Spectra
- EMRI Dynamics and Source-Driven Ringdown
- Gravitational-Wave Polarization and Gauge Invariants
- Rotating Wormholes and Wave Propagation
- Quantum Field Theory in Curved Spacetime
- Resolvent, Green-Function, and Spectral Methods
- Modified Gravity and Geometric Invariants

## Publications

### Peer-Reviewed Journal Publications

*Nine peer-reviewed publications; first-author papers are marked with ⋆.*

⋆ **R. Radhakrishnan** et al., *Where are linearized gauge invariants encoded for plane waves in linearized gravity*, **European Physical Journal C 86**, 1183 (2026). https://doi.org/10.1140/epjc/s10052-026-16355-y

⋆ **R. Radhakrishnan** et al., *Coupled-Channel Spectral Theory for a Non-Separable Geometry: Normal Modes and Two-Boundary Response in the Rotating AdS-Teo Wormhole*, **Symmetry 18**, 1497 (2026). https://doi.org/10.3390/sym18091497

⋆ **R. Radhakrishnan** et al., *Gauge-Invariant Gravitational Wave Polarization in Metric $f(R)$ Gravity with Cosmological Implications*, **Universe 12**, 44 (2026). https://doi.org/10.3390/universe12020044

1. C. Cremaschini, **R. Radhakrishnan** et al., *Invariant Boltzmann-Shannon Entropy for Black-Holes: A Manifestly-Covariant Canonical Quantum-Gravity Approach*, **Entropy 28**, 932 (2026). https://doi.org/10.3390/e28080932

2. C. Cremaschini, **R. Radhakrishnan** et al., *The Validity of the Ehrenfest Theorem in Quantum Gravity Theory*, **Symmetry 18**, 182 (2026). https://doi.org/10.3390/sym18010182

⋆ **R. Radhakrishnan** et al., *A Review of Stable, Traversable Wormholes in $f(R)$ Gravity Theories*, **Symmetry 16**, 1007 (2024). *(Review article.)* https://doi.org/10.3390/sym16081007

3. C. Brown, **R. Radhakrishnan** et al., *Killing Invariants: An Approach to the Sub-Classification of Geometries with Symmetry*, **General Relativity and Gravitation 56**, 92 (2024). https://doi.org/10.1007/s10714-024-03277-x

4. M. Gorban, **R. Radhakrishnan** et al., *First- and Second-Order Forces in the Asymmetric Dynamical Casimir Effect*, **Physics 6**, 760–779 (2024). https://www.mdpi.com/2624-8174/6/2/47

5. M. J. Gorban, W. D. Julius, **R. Radhakrishnan**, and G. B. Cleaver, *Interference phenomena in the asymmetric dynamical Casimir effect for a single mirror*, **Physical Review D 108**, 096037 (2023). `https://doi.org/10.1103/PhysRevD.108.096037`

### Manuscripts Being Prepared for Submission

- **R. Radhakrishnan** and G. Cleaver, *Hidden Self-Adjoint Structure, Exceptional Points, and Stability in Non-Reciprocal Coupled Oscillators*. Planned submission to **Axioms** (October 2026; not yet submitted).
- **R. Radhakrishnan** and G. Cleaver, *Connecting Quasinormal-Mode Splitting to Resolvent Residues at Black-Hole Exceptional Points*. Planned submission to **Axioms** (October 2026; not yet submitted).

### Research Manuscripts in Progress

- **R. Radhakrishnan** et al., *Particle-Driven Excitation and Suppression of Exceptional-Point Ringdown in Kerr Spacetime*. Massive-scalar Kerr source-excitation framework; causal waveform and physical suppression conditions under investigation.
- **R. Radhakrishnan** et al., *A Coupled Spectral Framework for Scalar Perturbations of the Rotating Teo Wormhole: Spectral Hybridization and Quasinormal Modes*. Manuscript undergoing revision / final review; prior submission reported. `https://arxiv.org/abs/2512.23104`

## Conference Presentations

- *Listening to a Rotating Wormhole: Quasinormal Modes and Time-Domain Ringdown*, 2026 Fall Texas, Arkansas, Oklahoma Section Meeting of the American Physical Society (TAOAPS), Corpus Christi, TX (2026). *Abstract submitted; presentation pending.*
- **Invited Talk:** *Near-Throat Conformal Structure and Global Mode Spectrum in the Rotating AdS–Teo Wormhole*, Troy University, Troy, AL (April 2026)
- **Invited Talk:** *Stationary Particle Creation and Entanglement in the Rotating Teo Wormhole: A Quantum Mode-Mixing Approach*, Troy University, Troy, AL (April 2026)
- **Invited Talk by Zoom:** *Stationary Particle Creation and Entanglement in the Rotating Teo Wormhole: A Quantum Mode-Mixing Approach*, Theoretical Physics Group, Raman Research Institute (RRI), Bangalore, India (April 2026)
- *Near-Throat Conformal Structure and Global Mode Spectrum in the Rotating AdS–Teo Wormhole*, 12th Gulf Coast Relativity Meeting, Troy, AL (April 2026)
- *Quantum Mode Mixing in the Teo Rotating Wormhole*, APS Global Physics Summit, Denver, CO (March 2026)
- *Quasinormal Modes in the Rotating Teo Wormhole*, Midwest Relativity Meeting, University of Guelph, ON (2025)
- *Gauge-Invariant Bardeen Variables for Plane Waves*, TAOAPS, Fort Worth, TX (2025)
- *Stable, Traversable Wormholes in $f(R)$ Gravity Theories*, Midwest Relativity Meeting, Ann Arbor, MI (2024)
- *Gauge-Invariant Bardeen Variables for Plane Waves*, APS April Meeting, Sacramento, CA (2024)
- *Stable, Traversable Wormholes in $f(R)$ Gravity*, Texas APS Meeting, SMU, Dallas, TX (2024)
- *Asymmetric Dynamical Casimir Effect*, Poster, APS April Meeting, Minneapolis, MN (2023)

## Research Experience

**Affiliate Research Professor — EUCOS Group, Baylor University** June 2026–Present

- **Exceptional points and black-hole response:** Developing analytical and numerical connections among Jordan chains, spectral splitting, resolvent poles, and source-driven excitation in Kerr and related black-hole perturbation models.
- **Rotating-spacetime wave dynamics:** Studying spectral hybridization, quasinormal modes, and time-domain response in rotating wormhole backgrounds.
- **Proposed postdoctoral direction:** Investigate EP sensitivity and source excitation along EMRI inspiral-to-plunge trajectories, with spin-2 perturbations and waveform consequences as longer-term goals.

**Graduate Research Assistant — EUCOS Group, Baylor University** 2021–2026

- Developed analytic and spectral treatments of scalar perturbations in rotating wormhole geometries, including coupled angular channels.
- Studied near-throat conformal structures, boundary response, and quantum mode mixing in rotating spacetimes.
- Investigated gauge-invariant gravitational-wave observables and geometric classification methods in general and modified gravity.
- Contributed calculations to collaborative research on quantum-gravity models, geometric invariants, and dynamical Casimir effects.

Advisor: **Prof. Gerald Cleaver**

**Graduate Research Assistant — Quantum & Nano Optics Lab** 2020–2021

- Modeled THz metamaterial structures using COMSOL Multiphysics.

Advisor: **Prof. David Hilton**

## Teaching Experience

**Graduate Prelim Prep Tutor - Statistical Mechanics, Electrodynamics and Quantum Mechanics** 2026

**Graduate Prelim Prep Instructor - Electrodynamics and Quantum Mechanics, Baylor University** 2025

- Led advanced review workshops in Electromagnetism, Statistical Mechanics and Quantum Mechanics

**AP Physics & AP Calculus Tutor** 2020–2022

- Mentored high school and undergraduate students in advanced STEM coursework.

## Technical Skills

- **Computing:** Python, Mathematica, Maple, C, NumPy, Pandas, Seaborn, SQL, Oracle, MySQL
- **Methods:** WKB, AIM, spectral methods, numerical GR, tensor algebra, differential geometry
- **Toolkits:** EinsteinPy, GRQUICK, COMSOL, Jupyter, Google Colab, PyCharm

## Awards & Fellowships

- **NASA Texas Space Grant Consortium Graduate Fellowship** 2025
- **2025-2026 Outstanding Dissertation**, Department of Physics and Astronomy, Baylor University 2026
- **Postdoctoral Teaching Fellowship Nomination**, Baylor University 2025
- Graduate School Fellowship & Tuition Scholarship, Baylor University 2020–2026
- APS DGRAV Travel Awards; Texas APS Student Ambassador Award 2023–2024
- LSU Gravity Workshop Travel Fellowship 2023

## Outreach and Mentorship

- **Volunteer Physics Mentor, National Society of Black Physicists (NSBP)** Fall 2026–Present
  Selected for the inaugural 2026 Uplifting Physicists Program (UPP) mentor cohort; mentoring a senior undergraduate physics student at the University of California, Los Angeles (UCLA), providing academic and professional guidance.

## Leadership & Service

- **Nominated Candidate, APS DGRAV Executive Committee Member-at-Large** 2026
- Student Ambassador, American Physical Society 2023–2024
- **Prior Career (1992–2022):** Led global engineering teams on multimillion-dollar projects.
- Member, Project Management Institute; Associate Member, Institution of Engineers (India).